SE Lab – 03

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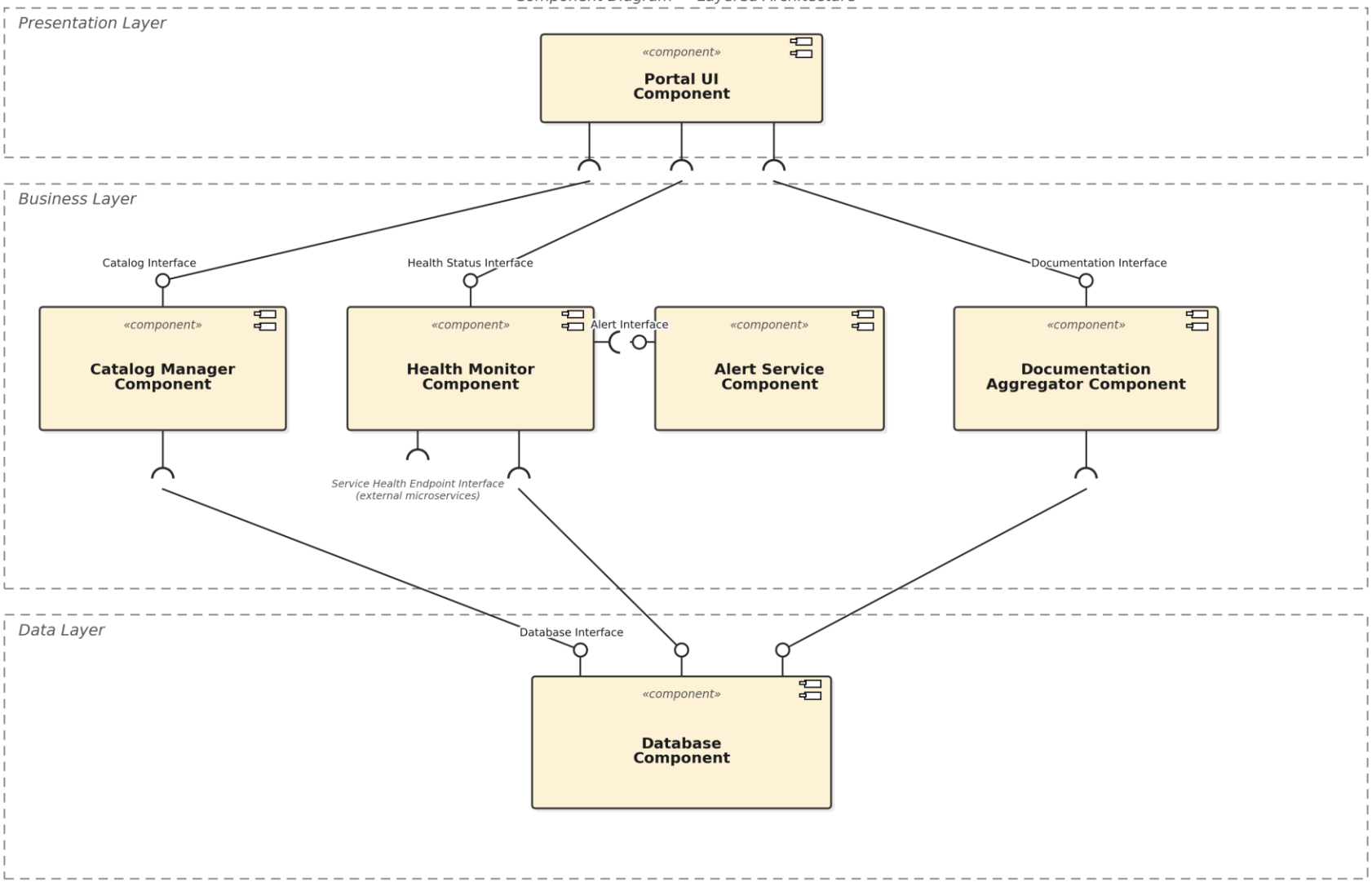
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SECTION : B

Lab 3 — Component Diagram: Internal Microservice Catalog & Health Portal

Internal Microservice Catalog & Health Portal

Component Diagram — Layered Architecture



Lab 3: Component Modelling & Architectural Pattern Selection

System: Internal Microservice Catalog & Health Portal

We chose Layered Architecture for the Internal Microservice Catalog & Health Portal System.

Architectural Choice

I split the portal into three layers. The Presentation Layer is just the Portal UI Component. Underneath it sits the Business Layer, holding the Catalog Manager, Health Monitor, Alert Service, and Documentation Aggregator. At the bottom is the Data Layer, a single Database Component that holds the catalog, the dependency links between services, the health-check history, and the documentation metadata.

Two Reasons for This Choice

1. The portal genuinely does five different jobs — keep the catalog current, draw the dependency graph, surface API docs, watch service health, and fire alerts (FR-001 through FR-005). Putting each job in its own layer/component means I can change how alerts work, say, without touching how the catalog is stored.

2. This is one internal tool running continuously for our own engineers, not a set of public services that need to scale independently — so microservices would add deployment and coordination overhead for no real benefit here. A layered setup is simpler to build and still handles the interactive 200-service dependency graph that NFR-001 asks for.

Security Advantage

The UI never talks to the database directly — only the Catalog Manager, Health Monitor, and Documentation Aggregator are allowed to use the Database Interface. That keeps every read and write going through one vetted layer instead of letting the user-facing component touch data directly, so there's a single, well-defined place to enforce access control.

Performance Benefit

Since the Health Monitor, Alert Service, and Database Component all live in the same layer, checking the rolling 24-hour availability and counting three consecutive failures (FR-001, FR-005) doesn't need to bounce up to the UI and back. The alert can go out within the same monitoring cycle, which is exactly what NFR-002 requires.